Trustworthy Machine Learning

Beyond PAC Learning

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Is PAC Learning Okay?

Four Ingredients of Learning:

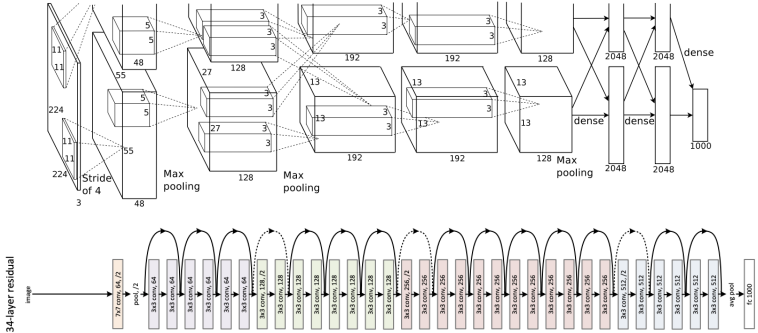
- Distribution $\mathcal{D}$
- Loss ℓ
- Hypothesis Space $\mathcal{H}$
- A Learning Algorithm $\mathcal{A}$

Problem?

The main assumption of PAC learning: $\mathcal{D}$ is separable by some $h^* \in \mathcal{H}$.

Generally $\mathcal{D}$ Is Not Separable

Usually we do not know a set of hypotheses $\mathcal{H}$ that has the true hypothesis h^*.



- What is the architecture of neural networks that perfectly classifies ImageNet?
- We mainly search for good hypothesis space $\mathcal{F}$ without any assumption on $\mathcal{D}$.

Contents

1 Concentration Inequalities

2 Generalization Bounds via Uniform Convergence

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Why Concentration Inequalities?

- Understanding the expected loss is a key in statistical learning

$$\min_{f \in \mathcal{F}} \mathbb{E}\ell(x, y, f)$$

- Concentration inequalities
  - Powerful tools that show an appropriate combination of independent random variables will *concentrate* around its *expectation*.
  - A concentration inequality provides a bound around an expected value.
- An Example: Mean estimation
  - Let $X_1, \ldots, X_n$ be i.i.d. real-valued random variables with mean $\mu := \mathbb{E}[X_1]$
  - The empirical mean is defined as

$$\hat{\mu}_n := \frac{1}{n} \sum_{i=1}^{n} X_i$$

  - What is the relation between μ and $\hat{\mu}_n$?

Possible Arguments I

Consistency: Due to the law of large numbers,

$$\hat{\mu}_n - \mu \xrightarrow{P} 0$$

- $\xrightarrow{P}$: convergence "in probability"
- If we get more data, $\hat{\mu}_n$ reaches to μ

✗ Asymptotic guarantee: it does not answer on the required number of samples to reach to the correct answer.

Possible Arguments II

Asymptotic normality: Assuming $\text{Var}(X_1) = \sigma^2$, due to the central limit theorem,

$$\sqrt{n}(\hat{\mu}_n - \mu) \xrightarrow{D} \mathcal{N}(0, \sigma^2)$$

- $\xrightarrow{D}$: convergence "in distribution"
- If we get more data, $\hat{\mu}_n$ reaches to μ, where the variance is decreasing at a rate of $1/n$.

✗ Asymptotic guarantee: it does not answer on the required number of samples to reach to the correct answer.

Possible Arguments III

Tail bound: we wish to have a statement as follows:

$$\mathbb{P}\left\{|\hat{\mu}_n - \mu| \geq \varepsilon\right\} \leq \text{SomeFunctionOf}(n, \varepsilon) = \delta.$$

- ε: a desired error level
- $1 - \delta$: the confidence of the error statement

✓ "SomeFunctionOf$(n, \varepsilon) = \delta$" provides the required number of samples to reach a desired level of error with a desired level of confidence.

Hoeffding's Inequality

Theorem

Let $X_1, \ldots, X_n$ be independent random variables with $X_i \in [a_i, b_i]$ for all $i \in \{1, \ldots, n\}$. Then, for any $\varepsilon > 0$, the following inequalities hold for $S_n := \sum_{i=1}^n X_i$:

$$\mathbb{P}\left\{\mathbb{E}\{S_n\} - S_n \geq \varepsilon\right\} \leq \exp\left\{\frac{-2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right\}$$

- Why is it called a tail bound?
- What's the effect of n?
- $X_1, \ldots, X_n$ need not to follow the same distribution

Binomial Distribution Tail Bound

A special version of the Hoeffding's inequality.

Theorem

Let $X_1, \ldots, X_n$ be independent random variables with $X_i \in \{0, 1\}$ and $\mathbb{P}\{X_i = 1\} = p \in [0, 1]$ for all $i \in \{1, \ldots, n\}$. Then, for any $\varepsilon > 0$, the following inequalities hold for $S_n = \sum_{i=1}^{n} X_i$:

$$\mathbb{P}\left\{p \leq \hat{p}\right\} \geq 1 - \delta,$$

where $F(k; n, p)$ is the CDF of a binomial distribution with n trials and success probability p and $\hat{p} := \inf\left\{p' \in [0, 1] \mid F(S_n; n, p') \leq \delta\right\}$.

- This is from the Clopper-Pearson interval for estimating binomial confidence intervals.
- $\hat{p}$ is the "best" value that fails to "describe" our observation S_n.
- From the Hoeffding's inequality, $\mathbb{P}\left\{\frac{S_n}{n} - p > \varepsilon\right\} \leq \exp\left\{-2n\varepsilon^2\right\}$

McDiarmid's Inequality

A generalized version of the Heoffding's inequality.

Theorem

Let $(X_1, \ldots, X_n) \in \mathcal{X}^n$ be a list of $n \geq 1$ independent random variables and assume that there exist $c_1, \ldots, c_n > 0$ such that $f : \mathcal{X}^n \to \mathbb{R}$ satisfies the following conditions:

$$\left| f(x_1, \ldots, x_i, \ldots, x_n) - f(x_1, \ldots, x_i', \ldots, x_m) \right| \leq c_i,$$

for all $i \in \{1, \ldots, n\}$ and any $x_1, \ldots, x_n, x_i \in \mathcal{X}$. Let $f(S)$ denote $f(X_1, \ldots, X_n)$, then, for all $\varepsilon > 0$, the following inequality holds:

$$\mathbb{P}\left\{\mathbb{E}\{f(S)\} - f(S) \geq \varepsilon\right\} \leq \exp\left\{\frac{-2\varepsilon^2}{\sum_{i=1}^n c_i^2}\right\}.$$

- The main inequality for a generalization bound.

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